

PrimeVarClass

Glossary



Support document - English

Translational research, computational validation, and responsible interpretation of missense variants.

PrimeVarClass - Glossary

Version: 2026.05

This glossary explains the most important platform terms in accessible language. It is intended for both beginners and experienced researchers.

ACMG/AMP

Guidelines used to interpret genetic variants. PrimeVarClass can help organize evidence, but it does not replace formal ACMG/AMP curation.

AlphaFold DB

Database of predicted protein structures. In PrimeVarClass, it helps locate where a variant occurs in the protein.

Limit: AlphaFold is structural prediction, not experimental structure.

AlphaMissense

External predictor for missense variant effects. It can be used as an independent comparison against PrimeVarClass.

Correct use: compare agreement and disagreement, not as final truth.

API

Programmable interface that allows other systems to communicate with the platform. PrimeVarClass uses a FastAPI backend.

API key

Access key used to protect the platform when exposed on a network.

Best practice: never publish the key in a paper, screenshot, repository, or shared document.

Baseline

Reference model or method used for comparison.

Example: a non-prime random forest, logistic regression, or external predictor.

Batch screening

Analysis of many variants at once, usually from a CSV file.

Benchmark

Evaluation designed to measure model performance under defined conditions.

Frozen benchmark

Benchmark where data, splits, parameters, thresholds, and metrics are fixed before evaluation.

Why it matters: reduces leakage, overfitting, and opportunistic interpretation.

BRCA Exchange

Specialized database for BRCA1 and BRCA2 variants. Useful for BRCA-specific validation and expert comparison.

cBioPortal

Platform containing tumor cohort data. In PrimeVarClass, it helps connect genes and variants to oncology context.

Limit: somatic tumor evidence should not automatically be treated as germline pathogenicity evidence.

ClinGen ERepo

Evidence repository containing expert-panel classifications. Important for external validation and independent curation.

ClinVar

Public database of genetic variants and clinical interpretations submitted by laboratories and expert groups.

Limit: ClinVar may include conflicts, different review levels, and older submissions.

Cohort

Set of variants, patients, samples, or records used in an analysis.

Examples: training cohort, external cohort, BRCA cohort, TP53 cohort.

Computational preprint

Manuscript stage where software, data processing, benchmarks, and evidence are strong enough for computational publication, but not necessarily for clinical claims.

Data leakage

When information from validation or test data indirectly enters training.

Risk: performance looks better than it truly is.

DFT

Density Functional Theory. Quantum chemistry method used to estimate energetic and electronic properties.

Use in PrimeVarClass: stronger follow-up after fast xTB screening.

Docking

Simulation of how a molecule may bind to a protein.

Limit: docking suggests a hypothesis, not therapeutic efficacy.

External holdout

Dataset kept separate from training and used to evaluate generalization.

Feature

Variable used by a model.

Examples: gnomAD frequency, MAVE score, conservation score, prime gap, prime curvature.

Fingerprint

Technical summary of an artifact, usually including hash, size, and date.

Use: reproducibility and audit.

Functional evidence

Evidence that measures or suggests the effect of a variant on biological function.

Examples: MAVE/DMS, cellular assay, transactivation, stability, enzymatic activity.

GDC

NCI Genomic Data Commons. Source of cancer genomic data.

Gene

DNA region that encodes or regulates a functional product. In the platform, gene is the main unit for grouping variants and evidence.

gnomAD

Genome Aggregation Database. Public source of population allele frequencies.

Interpretation: high frequency may argue against strong pathogenicity for rare disease; low frequency does not prove pathogenicity.

GWAS Catalog

Database of genetic associations between variants, regions, genes, and traits.

Use: biological plausibility and gene-disease context.

HGVS protein notation

Standardized protein variant notation.

Example: p.Cys61Gly means cysteine changed to glycine at position 61.

Independent validation

Evaluation on data not used during model development.

Interpretability

Ability to understand why a result was produced.

Examples: frequency, conservation, MAVE score, structural region, prime signature.

Label

Target class used for training or evaluation.

Examples: benign, pathogenic, VUS, functional, non-functional.

LOVD

Leiden Open Variation Database. A collection of locus-specific variant databases.

Manifest

JSON file recording paths, parameters, sources, versions, hashes, and generated artifacts.

Why it matters: enables reproducibility and audit.

MAVE

Multiplexed Assay of Variant Effect. High-throughput assay measuring effects of many variants.

MaveDB

Public repository of MAVE studies.

Use in PrimeVarClass: independent functional evidence.

Missense variant

Variant that changes one amino acid to another.

Example: p.Arg175His.

Multigene generalization

Ability of the method to work beyond one gene.

Example: expanding from BRCA1/BRCA2 to TP53, PTEN, MSH2, KRAS, GCK, and F9.

Non-prime control

Model, initialization, or feature set that does not use the prime-number layer.

Why it matters: tests whether prime numbers add real value.

Open Targets

Gene-disease evidence platform. Useful for translational context.

Overfitting

When a model learns training-specific details but fails on new data.

Warning sign: excellent training performance and weak external performance.

PDB

Protein Data Bank. Repository of experimental protein structures.

Use: strengthens structural interpretation when a relevant structure exists.

PharmGKB

Pharmacogenomic knowledge base. Helps contextualize genes and variants in relation to drugs and therapeutic response.

Preflight

Check performed before a main execution.

It detects missing files, incomplete schemas, leakage risk, insufficient data, and reproducibility problems.

Prime numbers

Numbers divisible only by 1 and themselves.

In PrimeVarClass, prime numbers are used to encode amino acids, substitutions, distances, and discrete patterns.

Scientific importance: the prime layer becomes credible only when it beats fair non-prime controls.

Prime-guided VQE

Use of prime-number signatures to guide initialization, fragment choice, active-space seed, or execution strategy in VQE.

Required control: compare with non-prime VQE initialization.

Protein structural context

Information about where a variant lies in a protein and what that region does.

Examples: domain, interface, catalytic site, disordered region.

Reproducibility

Ability to repeat an analysis with the same data, versions, and parameters.

Platform tools: manifests, fingerprints, reports, configs, and logs.

Shadow pilot

Pilot mode where the platform runs alongside the human process without influencing final decisions.

Use: measure safety, clarity, and usefulness before real adoption.

Split

Division of data into training, validation, test, or external cohorts.

Staging

Process of placing data in a standardized, traceable local format ready for use.

Threshold

Cutoff used to convert probability into a class.

Best practice: define it before external evaluation.

Translational impact

Evidence that the platform is useful in real scientific or laboratory workflows.

Examples: time saved, better prioritization, clearer evidence, safer review.

UniProt

Protein knowledge base providing accessions, length, domains, function, and annotations.

VQE

Variational Quantum Eigensolver. Hybrid quantum-classical algorithm for estimating energy in simplified systems.

VUS

Variant of Uncertain Significance.

Important: a VUS should not be treated as benign or pathogenic without additional evidence.

xTB

Fast semiempirical method for chemical and structural screening.

Use: first-pass triage before more expensive calculations such as DFT.

Actionability

Degree to which evidence can guide a practical research, experimental, translational, or therapeutic-investigation action.

Important: actionability in the platform is not an automatic clinical recommendation.

API key

Code used to authenticate a request or identify an authorized user.

Safety: never expose API keys in public screenshots, reports, repositories, or messages.

Artifact

Any file used or produced during an analysis.

Examples: trained model, report, manifest, log, chart, result table, configuration, and study package.

Authentication disabled

State in which the platform allows local use without requiring an API key.

Use: common in development, local validation, and controlled demonstrations.

Browser cache

Local storage used by a browser to load pages, styles, and scripts faster.

Common issue: after an interface update, the browser may show an older version. Refreshing with `Ctrl+F5` usually resolves it.

Codon

Sequence of three DNA or RNA bases that encodes one amino acid or a stop signal.

Relation to missense variants: a base change can alter a codon and replace the resulting amino acid.

Data sync

Process of fetching, updating, or registering data from public scientific sources.

Use: keeps the platform aligned with current external evidence.

Engine

Computational component that performs a specific task.

Examples: inference engine, structural engine, quantum engine, public-data sync engine, and benchmark engine.

Feedback

User-submitted record describing an error, question, suggestion, or usability issue.

Good feedback includes module, input, expected result, observed result, and error message.

File path

Local or relative address indicating where a file is stored.

Examples: model directory, output directory, manifest path, and dataset path.

Handoff

Organized delivery of data, results, or artifacts from one step to another person, module, or process.

Example: a package containing manifest, results, and logs delivered for independent validation.

Interpretable inference

Prediction accompanied by explanations, evidence, or signals that help the user understand why a model produced a result.

Important: interpretability does not make a result infallible.

Interpretable inference with real data

Analysis mode that combines model prediction with evidence from public sources or real datasets.

Scientific value: reduces dependence on synthetic examples and strengthens external credibility.

Job

Computational task executed by the platform.

Examples: predict variant, sync data, train model, run benchmark, generate PDF, and create manifest.

Job queue

List of tasks waiting to be executed or processed.

Use: important for multiuser platforms, long analyses, and automated routines.

Local environment

Mode in which the platform runs on the user's own machine or internal development environment.

Use: suitable for tests, demonstrations, early validation, and private work.

Local profile

Profile used in a local environment when full authentication is not enabled.

Use: useful for development and initial validation, but less robust than complete multiuser identity.

Module

Functional area of the interface dedicated to one part of the platform.

Examples: Home, Models, Prediction, Team, Public data, Studies, Science, Impact, and Operation.

Multiuser

Capability for multiple people to use the platform with separated profiles, permissions, logs, and context.

Important: requires authentication, audit trails, feedback, documentation, and governance.

Nitrogenous bases

Chemical components of DNA and RNA. In DNA, the main bases are adenine, thymine, cytosine, and guanine.

Relation to the platform: genetic variants start as base changes and may result in amino acid changes.

Nucleotide

Basic unit of DNA or RNA, composed of a nitrogenous base, sugar, and phosphate group.

Relation to variants: nucleotide changes can alter codons and proteins.

Output directory

Folder where the platform writes results.

Examples: reports, manifests, logs, metrics, figures, and analysis packages.

Panel

Visual block in the interface that groups related information or actions.

Examples: status panel, result panel, team panel, and public-data panel.

Public catalog

Public scientific data source used for query, training, validation, or evidence enrichment.

Examples: ClinVar, gnomAD, MaveDB, UniProt, PDB, AlphaFold DB, Open Targets, PharmGKB, and GWAS Catalog.

Public data

Data made available by institutions, consortia, repositories, or scientific databases.

Important: public does not mean perfect. Every source has scope, version, bias, criteria, and limitations.

Request

Message sent by the interface or another system asking the API to perform an action.

Examples: query model, predict variant, download document, or start data sync.

Score

Numerical value produced by a model, database, or computational method.

Examples: model probability, MAVE functional score, allele frequency, conservation score, and estimated energy.

Study package

Organized set of files needed to review or reproduce an analysis.

Examples: configuration, manifest, results, logs, metrics, figures, and report.

Team ID

Identifier associated with the active team.

Use: relevant for collaboration, auditing, governance, and multiuser operation.

Web readiness

Degree to which the platform is prepared for external users over the web.

Includes: authentication, stability, documentation, security, feedback, logs, accessibility, and infrastructure.

Workbench

Main platform working area, available at /workbench.

Use: gathers the scientific and operational modules in a single interface.